FUNDAMENTAL ASTRONOMY

Spherical geometry and astronomical coordinates

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Academic revision · November 9, 2026

1 INTRODUCTION

Spherical astronomy describes directions on the sky and their dependence on an observer's location and time. This handout develops coordinate rotations, derives spherical-triangle relations, and applies them to horizontal and equatorial coordinates. Familiarity with elementary trigonometry and Cartesian coordinates is assumed.

Background reading: H. Karttunen et al., *Fundamental Astronomy*. Coordinate definitions were checked against the U.S. Naval Observatory's Sidereal Time and Altitude and Azimuth. USNO uses north-through-east azimuth; this handout uses south-through-west. These are student learning notes, not original research.

2 SPHERICAL TRIANGLES

Definition 2.1 (Sphere). A sphere of radius r and centre O is the set of points at distance r from O.

Definition 2.2 (Great and small circles). A plane through O intersects the sphere in a *great circle*, of radius r. A plane that intersects the sphere but does not pass through O gives a *small circle*. For non-antipodal endpoints, the shorter great-circle arc is the shortest path on the sphere.

Definition 2.3 (Poles). The line through O perpendicular to a great circle's plane meets the sphere at its two poles, P and P′. Every point of the great circle is 90° from either pole. Parallel small circles have the same axis and poles.

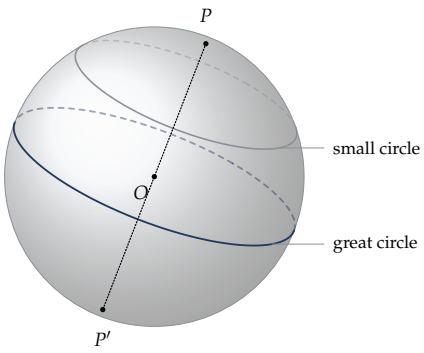


Figure 1: A great circle, a parallel small circle, and their poles P and P′ on the common perpendicular axis through O.

Definition 2.4 (Spherical triangle). Three distinct, pairwise non-antipodal points A, B, C, not on one great circle, determine a triangle bounded by shorter great-circle arcs. Its sides a = BC, b = CA and c = AB are angular lengths, opposite the corresponding vertices. An interior angle is measured between the tangents to the two arcs at a vertex. We use the convex triangle, with sides and angles between 0° and 180°.

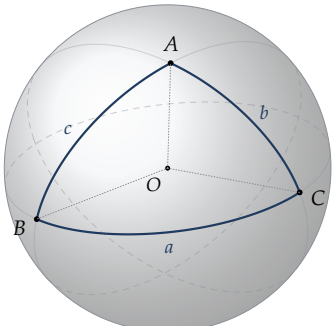


Figure 2: The triangle ABC: a, b and c are arcs of great circles, rather than straight segments in a plane.

On a sphere of radius r, an arc with central angle c has length

$$|AB| = rc, \tag{1}$$

Here c must be in radians. On a unit sphere, arc length and central angle have the same numerical value. When using degrees, convert to radians before computing a physical length. For a nondegenerate convex spherical triangle, 180° < A + B + C < 540°.

3 TRANSFORMATIONS

Place Cartesian axes at the centre of a unit sphere. Rotate the axes about x by χ in the sense that carries +y toward +z, giving x′ = x. The point P remains fixed: this is a *passive rotation*. We seek the coordinates of the same point in both frames.

Definition 3.1 (Angular coordinates). The longitude-like angle ψ is measured in the xy plane from +x toward +y, counterclockwise as viewed from +z, with $0° \le \psi < 360°$. The elevation θ is measured from that plane toward +z, with $-90° \le \theta \le 90°$. Define ψ' and θ' analogously in the rotated frame. At $\theta = \pm 90°$, ψ is undefined.

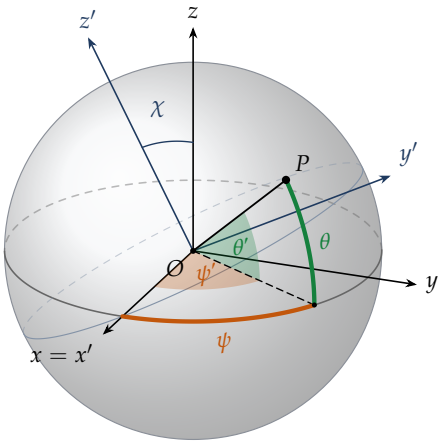


Figure 3: The fixed point P in the original and rotated frames. The inner angular labels are ψ' and θ'; the outer labels are ψ and θ.

3.1 The components of P

Project P perpendicularly onto the xy plane at Q. The right triangle OQP gives $z = \sin\theta$ and $OQ = r_{xy} = \cos\theta$. Resolving OQ in the xy plane then gives

$$x = \cos\theta\cos\psi, \qquad y = \cos\theta\sin\psi, \qquad z = \sin\theta. \tag{2}$$

The sketch shows acute angles. The same expressions hold over the full stated ranges because sine and cosine supply the correct signs.

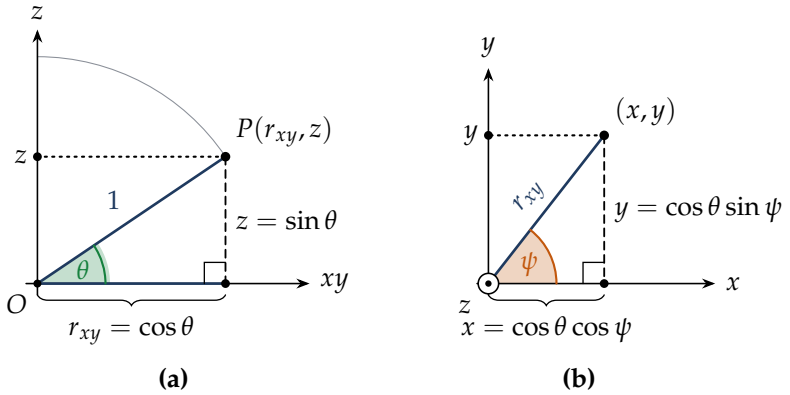


Figure 4: (a) Side view in the plane containing z and OQ. (b) Top view of the xy plane. The line labelled xy represents the projection OQ, not an additional Cartesian axis.

3.2 The rotation

Only y and z change. Projection onto y′ gives a contribution y cos χ from y and z sin χ from z:

$$y' = y\cos\chi + z\sin\chi. \tag{3}$$

Projection onto z′ gives z cos χ from z and −y sin χ from y, because +z′ tilts toward −y:

$$z' = z\cos\chi - y\sin\chi. \tag{4}$$

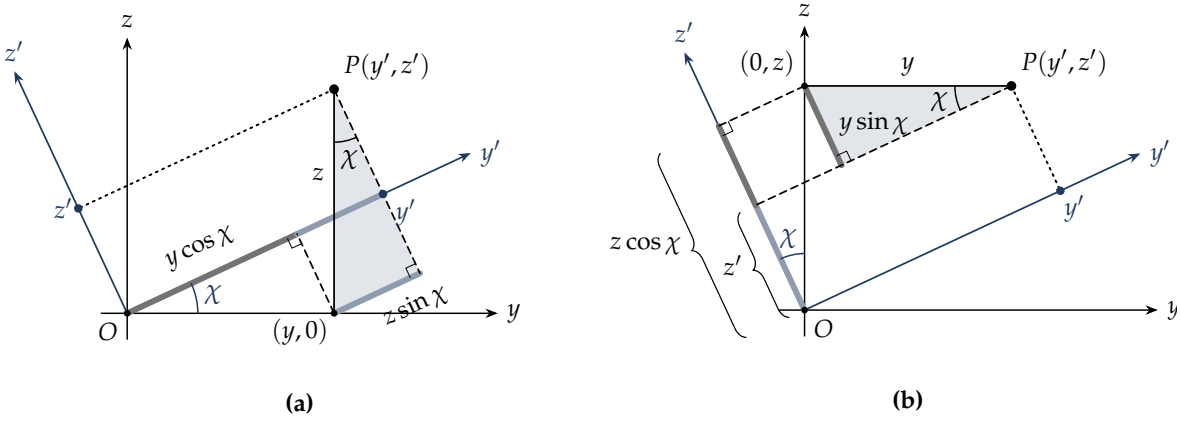


Figure 5: Orthogonal projections in the yz plane. (a) y′ = y cos χ + z sin χ. (b) z′ = z cos χ − y sin χ. The original two panels are placed side by side.

The inverse is obtained by replacing χ with −χ: y = y′ cos χ − z′ sin χ and z = y′ sin χ + z′ cos χ. An active rotation of the point by +χ, with axes fixed, uses the opposite sign convention.

3.3 The transformation equations

Substitute (2) and its primed version into x′ = x, (3) and (4):

Proposition 3.2 (Transformation equations).

$$\cos\psi'\cos\theta' = \cos\psi\cos\theta, \tag{5}$$

$$\sin\psi'\cos\theta' = \sin\psi\cos\theta\cos\chi + \sin\theta\sin\chi, \tag{6}$$

$$\sin\theta' = -\sin\psi\cos\theta\sin\chi + \sin\theta\cos\chi. \tag{7}$$

4 DERIVING THE SPHERICAL TRIANGLE EQUATIONS

Let A and B be the intersections of the +z and +z′ axes with the sphere, and rename P as C. The axis separation is the arc c = χ. Join A, B and C by shorter great-circle arcs (Fig. 6).

Definition 4.1 (The triangle ABC). A is the pole of the xy plane and B the pole of the x′y′ plane. Its sides are a = BC, b = CA and c = AB. Assume 0° < χ < 180° and a nondegenerate triangle.

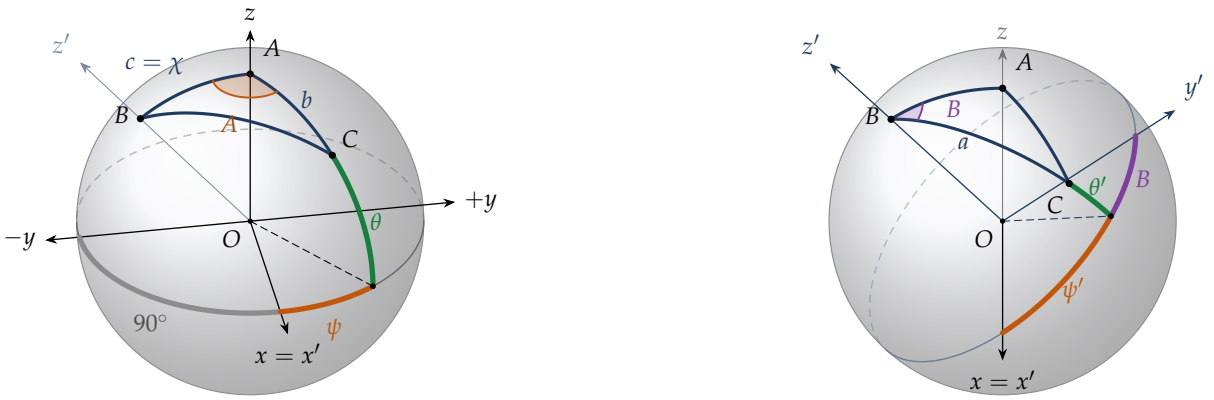


Figure 6: Relations between the triangle and the two angular coordinate systems. The configuration shown has C on the +x side of the yz plane.

Since a great circle is 90° from its poles, b + θ = 90° and a + θ′ = 90°. At a pole, the angle between two meridians equals the equatorial arc between them. For the orientation shown, A = 90° + ψ and B = 90° − ψ′. Thus

$$\theta = 90^\circ - b, \qquad \theta' = 90^\circ - a, \qquad \psi = A - 90^\circ, \qquad \psi' = 90^\circ - B. \tag{8}$$

Using c = χ, substitute cos ψ′ = sin B, sin ψ′ = cos B, cos θ′ = sin a and sin θ′ = cos a. Likewise cos ψ = sin A, sin ψ = −cos A, cos θ = sin b and sin θ = cos b. Equations (5)–(7) become

Theorem 4.2 (Spherical triangle equations).

$$\sin B \sin a = \sin A \sin b, \tag{9}$$

$$\cos B \sin a = -\cos A \sin b \cos c + \cos b \sin c, \tag{10}$$

$$\cos a = \cos A \sin b \sin c + \cos b \cos c. \tag{11}$$

These are the sine, sine–cosine and cosine relations. Cyclic relabelling gives the other forms. For the reflected triangle, the orientation changes but the identities remain valid for its interior angles. Degenerate cases, including three vertices on one great circle, require separate treatment.

5 ASTRONOMICAL COORDINATE SYSTEMS

We now apply the geometry to directions on the celestial sphere. In this section a, A and z mean altitude, azimuth and zenith distance; they no longer denote the triangle elements or Cartesian z used above.

5.1 The Earth

Use an ideal spherical Earth. Its equatorial plane is perpendicular to its rotation axis. Meridians run from pole to pole; the zero meridian is the conventional Greenwich meridian. Parallels of latitude are circles parallel to the equator.

Definition 5.1 (Longitude and latitude). Longitude λ is measured along the equator from the zero meridian to the observer's meridian, positive east and negative west, with $-180° < \lambda \le 180°$. Latitude φ is the angle of the local vertical above the equatorial plane, positive north, with $-90° \le \varphi \le 90°$. On a spherical Earth the local vertical is radial; the ellipsoidal Earth requires more precise latitude definitions.

For a northern observer, the north celestial pole's geometric altitude equals φ. Polaris is close to that pole, so its altitude only approximates latitude. In the south, the south celestial pole has altitude $|\varphi|$. Refraction and the difference between Polaris and the pole are omitted in this geometric model.

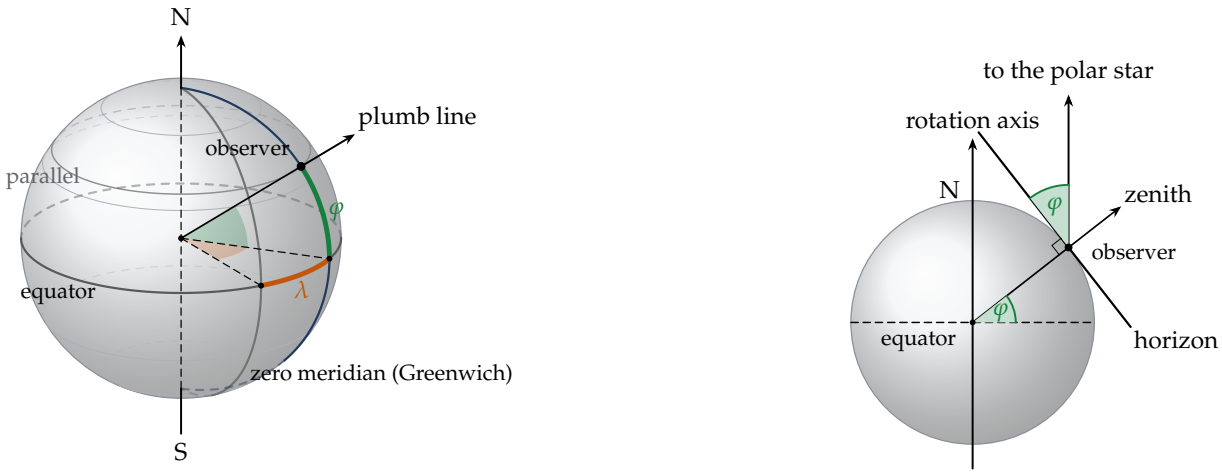


Figure 7: (a) Geographic longitude and latitude. (b) The polar direction is parallel to the rotation axis. Read the idealized "to the polar star" ray as the direction to the celestial pole; equality with latitude is exact for the pole, approximate for Polaris.

5.2 The horizontal system

The celestial sphere represents directions at an arbitrary radius about the observer. Extending Earth's rotation axis and equatorial plane defines the celestial poles and celestial equator. For distant stars, changing the centre from observer to Earth is negligible in this model; nearby objects may require parallax corrections.

Definition 5.2 (Horizontal coordinates). The reference plane is the horizon through the observer, perpendicular to the local vertical. Altitude a is measured from the horizon along the object's vertical circle, with $-90° \le a \le 90°$. Azimuth A increases from south through west, north and east, with $0° \le A < 360°$. In the north-through-east convention, $A_N = A + 180° \pmod{360°}$.

The zenith Z lies directly overhead; zenith distance is $z = 90° - a$. Azimuth is undefined at the zenith and nadir. Horizontal coordinates depend on location and time as Earth rotates.

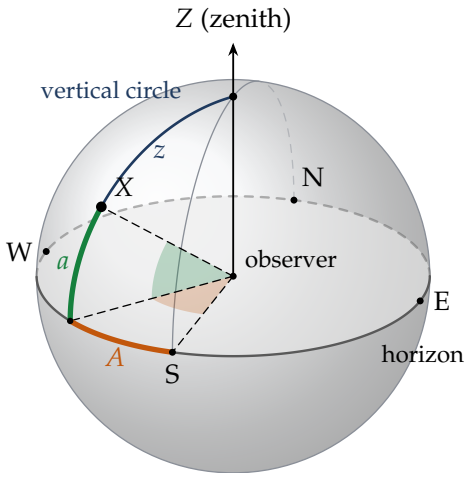


Figure 8: The horizontal system: azimuth A runs from S toward W, altitude a is above the horizon, and z is measured from the zenith Z.

5.3 The equatorial system

Definition 5.3 (Equatorial coordinates). Declination δ is the angular distance north (+) or south (−) of the celestial equator, with $-90° \le \delta \le 90°$. Right ascension α is measured eastward along the equator from the vernal equinox to the object's hour circle, with $0\mathrm{h} \le \alpha < 24\mathrm{h}$. One hour of right ascension is 15°. The hour circle passes through the object and both celestial poles.

The ecliptic is the Sun's apparent annual path. Of its two intersections with the equator, the vernal equinox is the northward crossing. Hour angle h is measured westward along the equator from the observer's meridian to the object's hour circle. At a fixed epoch, α and δ are approximately fixed for distant stars, whereas h changes with Earth's rotation.

Local sidereal time Θ is the hour angle of the vernal equinox. With α and h in consistent units,

$$\Theta = \alpha + h. \tag{12}$$

Equation (12) is understood modulo 24h (or 360°). Equivalently, $\mathrm{h} = \Theta - \alpha$. Greenwich and local sidereal times differ by east-positive longitude, expressed in the same angular or time units. At the celestial poles, right ascension and hour angle are undefined.

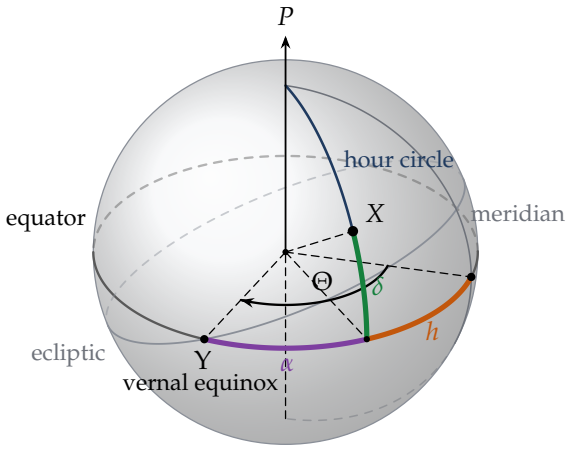


Figure 9: The equatorial system. Right ascension α increases eastward; hour angle h and local sidereal time Θ increase westward. Their sum relation is modulo a full turn.

5.4 From one system to the other

Choose a shared x axis pointing west. In the horizontal frame, y points south and z to the zenith. In the equatorial frame, y′ points toward the meridian's upper crossing of the equator and z′ to the north celestial pole. These right-handed frames differ by $\chi = 90^\circ - \varphi$. The angle definitions give

$$\psi = 90^\circ - A, \qquad \theta = a, \qquad \psi' = 90^\circ - h, \qquad \theta' = \delta, \qquad \chi = 90^\circ - \varphi. \tag{13}$$

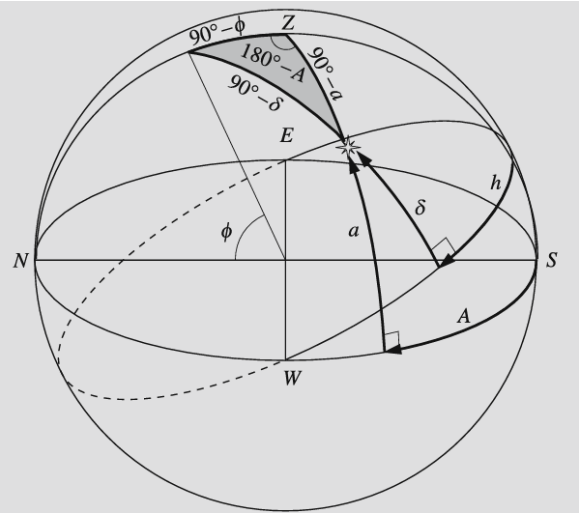


Figure 10: The pole–zenith–object triangle. Its sides are 90° − φ, 90° − δ and 90° − a. At Z, the interior angle is |180° − A|; at the pole it is the corresponding interior value of h.

Substituting (13) into (5)–(7) gives the horizontal-to-equatorial relations. Replacing χ by $-\chi$ gives the inverse. Keep all trigonometric arguments in the same angular unit; convert hours to degrees or radians first.

Theorem 5.4 (Horizontal to equatorial).

$$\sin h \cos \delta = \sin A \cos a, \tag{14}$$

$$\cos h \cos \delta = \cos A \cos a \sin \varphi + \sin a \cos \varphi, \tag{15}$$

$$\sin \delta = -\cos A \cos a \cos \varphi + \sin a \sin \varphi. \tag{16}$$

From equatorial to horizontal:

$$\sin A \cos a = \sin h \cos \delta, \tag{17}$$

$$\cos A \cos a = \cos h \cos \delta \sin \varphi - \sin \delta \cos \varphi, \tag{18}$$

$$\sin a = \cos h \cos \delta \cos \varphi + \sin \delta \sin \varphi. \tag{19}$$

As a check, the cosine formula (11), applied to the pole–Z–object triangle with angle h at the pole, gives (19). The relations hold in both hemispheres; for $\varphi < 0$ the north celestial pole is below the horizon.

For computation, obtain h or A from its sine *and* cosine using atan2, rather than inverse sine alone. For example, h = atan2(sin A cos a, cos A cos a sin φ + sin a cos φ). Then $\alpha = \Theta - h$, reduced modulo a full turn. Altitude follows from (19) in [−90°, 90°]. The meridian/south reference becomes degenerate at Earth's poles.